

2. Development Of Hello World Application.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:textSize="24sp"/>

</LinearLayout>
```

MainActivity.java

Code:

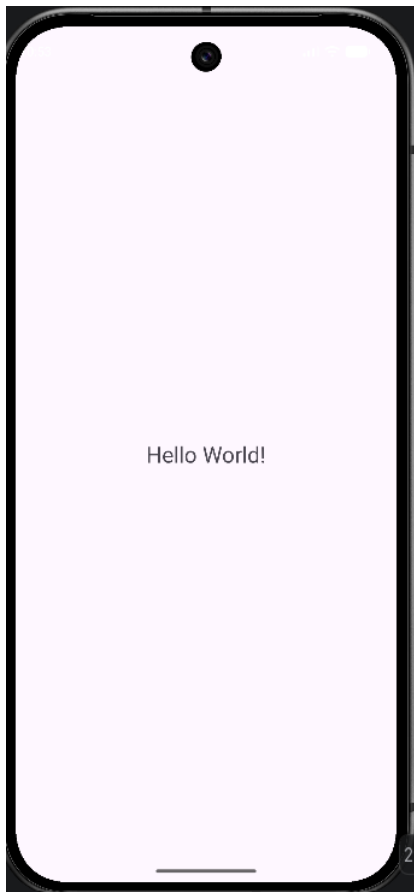
```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:



3. Create an application that takes the name from a textbox and shows hello message along with the name entered in text box, when the user clicks the OK button.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="50dp">

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name" />

    <Button
        android:id="@+id/ok"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="OK" />

    <TextView
        android:id="@+id/message"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"
        android:padding="20dp"/>

</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.name;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
```

```

import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText name;
    Button ok;
    TextView message;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        name = findViewById(R.id.name);
        ok = findViewById(R.id.ok);
        message = findViewById(R.id.message);

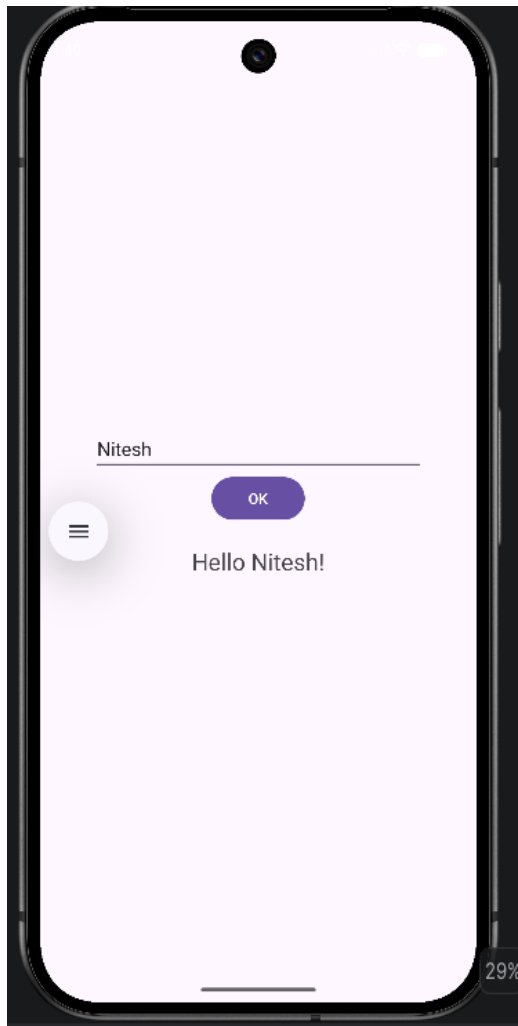
        ok.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String n = name.getText().toString();

                if (n.isEmpty()) {
                    message.setText("Please enter your name");
                } else {
                    message.setText("Hello " + n + "!");
                }
            }
        });
    }
}

```

Output:



4. Create a screen that has input boxes for User Name, Password, and Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (DatePicker), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button (use any layout).

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="50dp">

        <EditText
            android:id="@+id/username"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="User Name"/>

        <EditText
            android:id="@+id/password"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword"/>

        <EditText
            android:id="@+id/address"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Address"/>

        <RadioGroup
            android:id="@+id/gender"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <RadioButton
```

```
        android:id="@+id/male"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"/>

        <RadioButton
            android:id="@+id/female"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"/>
    </RadioGroup>

    <EditText
        android:id="@+id/age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Age"
        android:inputType="number"/>

    <Button
        android:id="@+id/dob"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Date of Birth"/>

    <Spinner
        android:id="@+id/state"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/submit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"/>

    <TextView
        android:id="@+id/output"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:padding="10dp"/>

</LinearLayout>
</ScrollView>
```

MainActivity.java

Code:

```
package com.example.formapp;

import android.app.DatePickerDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    EditText username, password, address, age;
    RadioGroup gender;
    Button dob, submit;
    Spinner state;
    TextView output;

    String date = "Not Selected";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        address = findViewById(R.id.address);
        age = findViewById(R.id.age);
        gender = findViewById(R.id.gender);
        dob = findViewById(R.id.dob);
        state = findViewById(R.id.state);
        submit = findViewById(R.id.submit);
        output = findViewById(R.id.output);

        String[] states = {"Maharashtra", "Gujarat", "Goa", "Delhi"};

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_spinner_dropdown_item,
            states);
```

```

state.setAdapter(adapter);

dob.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        Calendar c = Calendar.getInstance();

        new DatePickerDialog(
            MainActivity.this,
            (view, year, month, day) -> {

                date = day + "/" + (month + 1) + "/" + year;

                dob.setText(date);
            },
            c.get(Calendar.YEAR),
            c.get(Calendar.MONTH),
            c.get(Calendar.DAY_OF_MONTH)
        ).show();
    }
});

submit.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        int id = gender.getCheckedRadioButtonId();

        if (id == -1) {
            Toast.makeText(MainActivity.this,
                "Select Gender", Toast.LENGTH_SHORT).show();
            return;
        }

        RadioButton r = findViewById(id);

        String data =
            "User Name: " + username.getText().toString() +
            "\nPassword: " + password.getText().toString() +
            "\nAddress: " + address.getText().toString() +
            "\nGender: " + r.getText().toString() +
            "\nAge: " + age.getText().toString() +
            "\nDOB: " + date +
            "\nState: " + state.getSelectedItem().toString();
    }
});

```

```

        output.setText(data);
    }
});
}
}

```

Output:

Nitesh Vasave

SP Row House, Amrutdham, Nashik

☒ Male ☐ Female

25

22/2/2001

Maharashtra

Submit

User Name: Nitesh Vasave
 Password: Pass@123
 Address: SP Row House,
 Amrutdham, Nashik
 Gender: Male
 Age: 25
 DOB: 22/2/2001
 State: Maharashtra

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5. Design an android application to create page using Intent and one Button and pass the Values from one Activity to second Activity.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="50dp">

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <Button
        android:id="@+id/send"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Values"/>

</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.intentexp;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText name;
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    name = findViewById(R.id.name);

    Button sendButton = findViewById(R.id.send);
    sendButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent i = new Intent(MainActivity.this, SecondActivity.class);
            i.putExtra("name", name.getText().toString());
            startActivity(i);
        }
    });
}
}

```

AndroidManifest.xml

Code:

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.IntentExp">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>

```

```

        </intent-filter>
    </activity>

    <activity
        android:name=".SecondActivity"
        android:exported="false" />

</application>
</manifest>

```

activity_second.xml

Code:

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24sp"/>

</LinearLayout>

```

SecondActivity.java

Code:

```

package com.example.intentexp;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

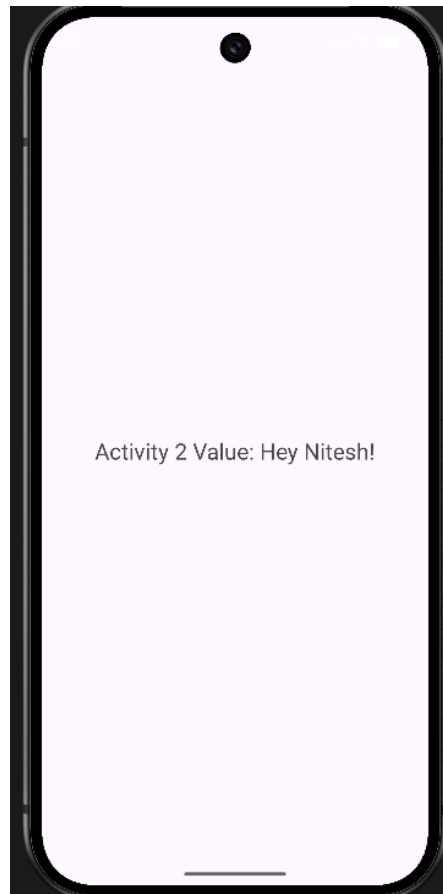
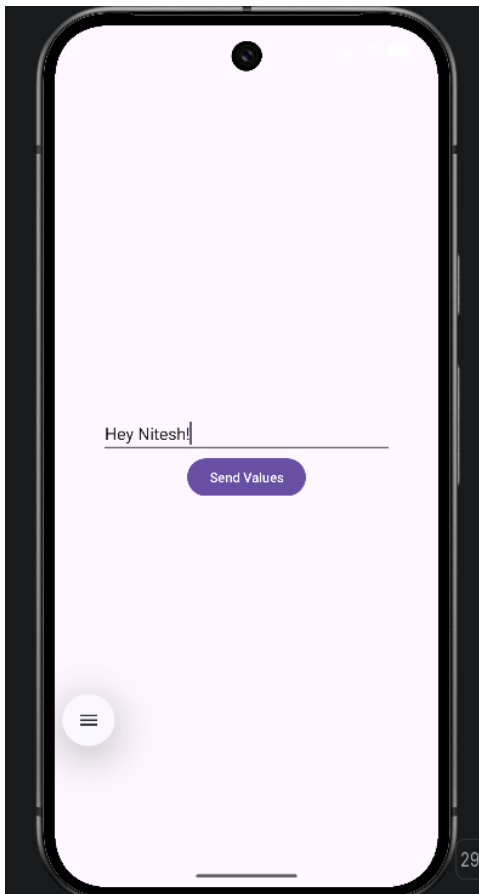
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        TextView result = findViewById(R.id.result);
    }
}

```

```
String name = getIntent().getStringExtra("name");  
result.setText("Activity 2 Value: " + name);  
}  
}
```

Output:



6. Design an android application Send SMS using Intent.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:layout_gravity="center"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Phone Number"
        android:inputType="phone"/>

    <EditText
        android:id="@+id/message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"/>

    <Button
        android:id="@+id/send"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send SMS"/>

</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.sms;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
```

```

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText phone, message;
    Button send;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phone = findViewById(R.id.phone);
        message = findViewById(R.id.message);
        send = findViewById(R.id.send);

        send.setOnClickListener(v -> {

            Uri uri = Uri.parse("smsto:" +
                phone.getText().toString());

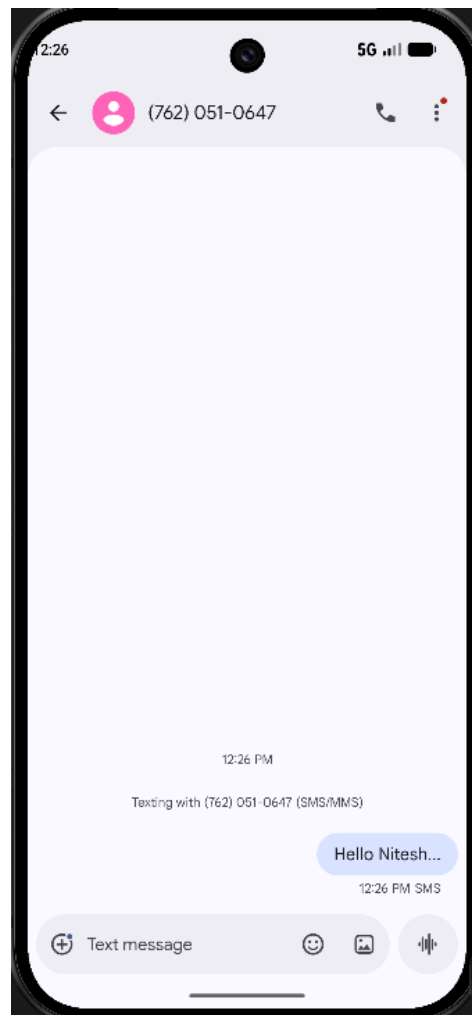
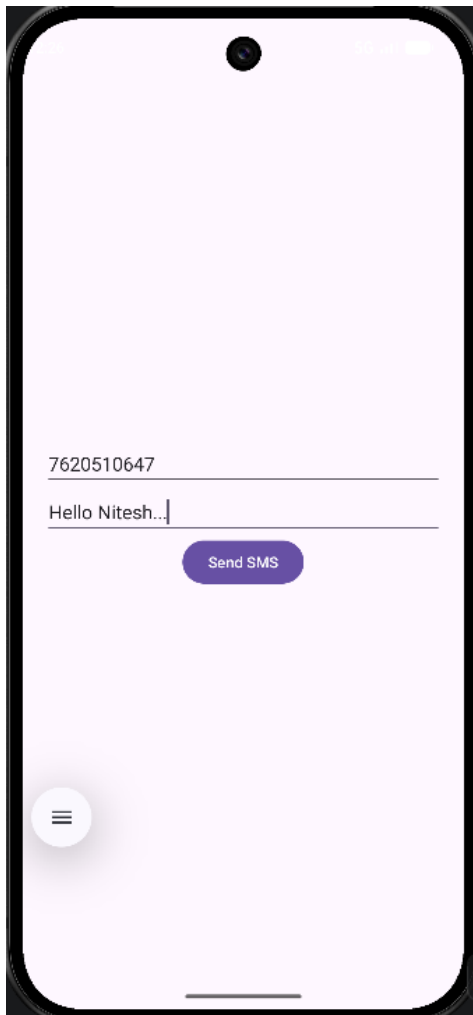
            Intent i = new Intent(Intent.ACTION_SENDTO, uri);

            i.putExtra("sms_body",
                message.getText().toString());

            startActivity(i);
        });
    }
}

```

Output:



7. Create an android application using Fragments.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:padding="8dp"
    android:background="#FFFFFF">

    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/list"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"/>

    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/detail"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="2"
        android:layout_marginStart="8dp"/>

</LinearLayout>
```

fragment_list.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="#E0F7FA">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SUBJECTS"
```

```
        android:textSize="18sp"
        android:textStyle="bold"
        android:textColor="#006064"
        android:gravity="center"
        android:layout_marginBottom="20dp"/>
```

```
<Button
    android:id="@+id/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

    android:text="Android"
    android:textSize="16sp"/>
```

```
<Button
    android:id="@+id/java"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Java"
    android:textSize="16sp"
    android:layout_marginTop="12dp"/>
```

```
</LinearLayout>
```

fragment_detail.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFF3E0">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:id="@+id/detail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Select a subject"
            android:textSize="25sp"
            android:textStyle="bold"
```

```

        android:textColor="#E65100"
        android:gravity="center"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="25dp"/>

<TextView
    android:id="@+id/info"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Select a subject from the left panel."
    android:textSize="17sp"
    android:textColor="#333333"
    android:gravity="center"
    android:lineSpacingExtra="6dp"/>

</LinearLayout>
</ScrollView>

```

DetailFragment.java

Code:

```

package com.example.fragapp;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.fragment.app.Fragment;

public class DetailFragment extends Fragment {

    TextView detail, info;

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
        savedInstanceState) {

        View v = inflater.inflate(R.layout.fragment_detail, container, false);
        detail = v.findViewById(R.id.detail);
        info = v.findViewById(R.id.info);

        return v;
    }

    public void show(String title, String text) {
        detail.setText(title);
        info.setText(text);
    }
}

```

```
}  
}
```

ListFragment.java

Code:

```
package com.example.fragapp;  
  
import android.content.Context;  
import android.os.Bundle;  
import android.view.LayoutInflater;  
import android.view.View;  
import android.view.ViewGroup;  
import android.widget.Button;  
import androidx.fragment.app.Fragment;  
  
public class ListFragment extends Fragment {  
  
    interface Listener {  
        void select(String item);  
    }  
  
    Listener listener;  
  
    @Override  
    public void onAttach(Context context) {  
        super.onAttach(context);  
        listener = (Listener) context;  
    }  
  
    @Override  
    public View onCreateView(LayoutInflater inflater, ViewGroup container,  
                             Bundle savedInstanceState) {  
  
        View v = inflater.inflate( R.layout.fragment_list, container, false);  
  
        Button android = v.findViewById(R.id.android);  
        Button java = v.findViewById(R.id.java);  
  
        android.setOnClickListener(x ->listener.select("Android"));  
        java.setOnClickListener(x -> listener.select("Java"));  
        return v;  
    }  
}
```

MainActivity.java

Code:

```
package com.example.fragapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
    implements ListFragment.Listener {

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        getSupportFragmentManager()
            .beginTransaction()
            .replace(R.id.list, new ListFragment())
            .replace(R.id.detail, new DetailFragment())
            .commit();
    }

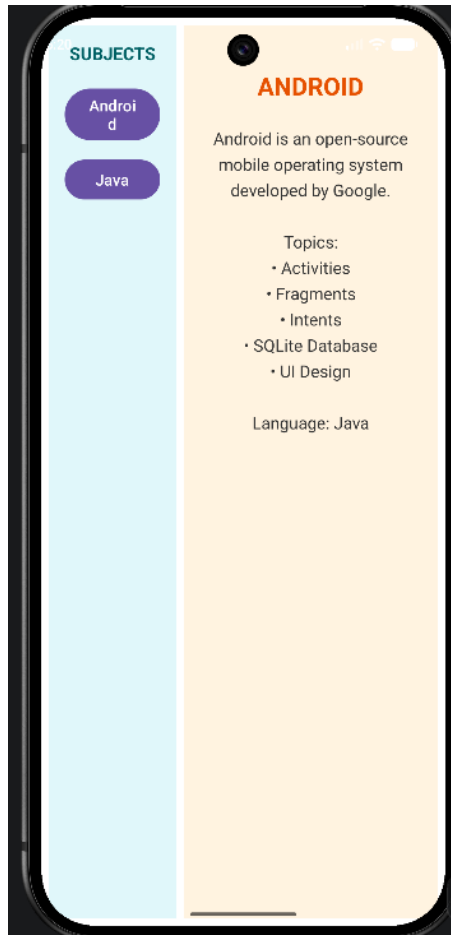
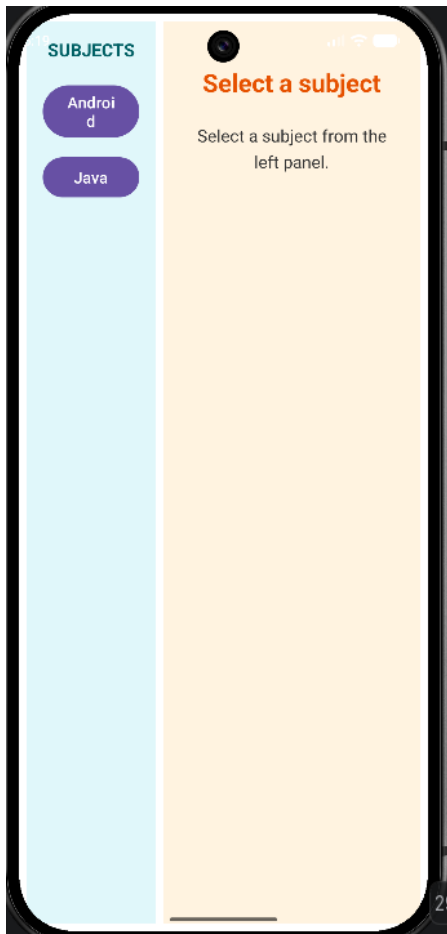
    @Override
    public void select(String item) {

        DetailFragment d = (DetailFragment)
            getSupportFragmentManager().findFragmentById(R.id.detail);

        if (d == null) return;

        if (item.equals("Android")) {
            d.show( "ANDROID",
                "Android is an open-source mobile " + "operating system developed by Google.\n\n"
                + "Topics:\n" + "• Activities\n" + "• Fragments\n" + "• Intents\n" +
                "• SQLite Database\n" + "• UI Design\n\n" + "Language: Java"
            );
        } else {
            d.show(
                "JAVA", "Java is a class-based, object-oriented " +
                "and platform-independent programming language.\n\n" + "Topics:\n"
                + "• Classes and Objects\n" + "• Inheritance\n" + "• Polymorphism\n" +
                "• Exception Handling\n" + "• Collections"
            );
        }
    }
}
```

Output:



8. Design an android application Using Radio buttons.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center_horizontal"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Subject"
        android:textSize="25sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:gravity="center"
        android:layout_marginTop="30dp"
        android:layout_marginBottom="25dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Choose your preferred subject:"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="15dp"/>

    <RadioGroup
        android:id="@+id/group"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <RadioButton
            android:id="@+id/java"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Java Programming"
            android:textSize="17sp"/>
```

```
<RadioButton
    android:id="@+id/network"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Computer Networking"
    android:textSize="17sp"/>
```

```
<RadioButton
    android:id="@+id/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Android Development"
    android:textSize="17sp"/>
```

```
</RadioGroup>
```

```
<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    android:layout_marginTop="25dp"/>
```

```
<TextView
    android:id="@+id/result"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Your selection will appear here."
    android:textSize="23sp"
    android:textStyle="italic"
    android:textColor="#2E7D32"
    android:gravity="center"
    android:layout_marginTop="30dp"/>
```

```
</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.radiobutton;

import android.os.Bundle;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;
```

```

public class MainActivity extends AppCompatActivity {

    RadioGroup group;
    Button submit;
    TextView result;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        group = findViewById(R.id.group);
        submit = findViewById(R.id.submit);
        result = findViewById(R.id.result);

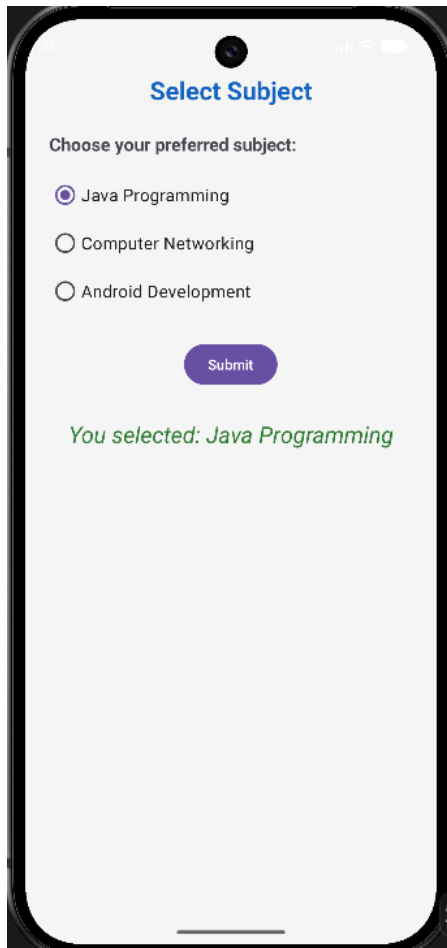
        submit.setOnClickListener(v -> {

            int id = group.getCheckedRadioButtonId();

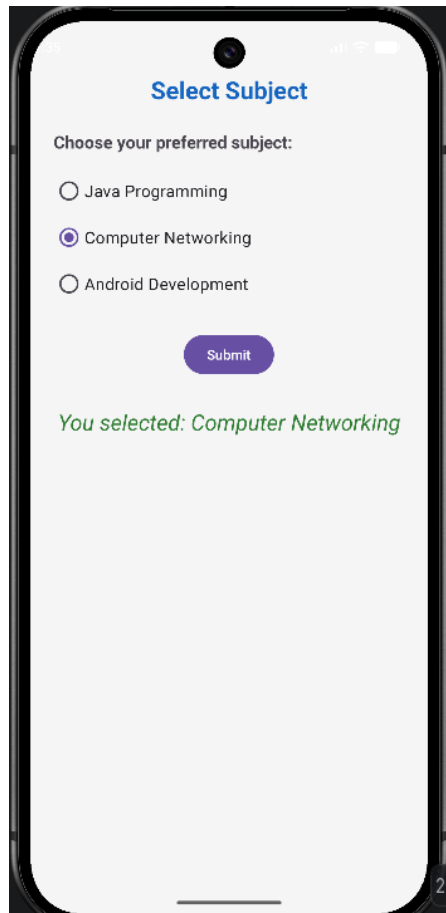
            if (id == -1) {
                result.setText("Please select a subject");
            } else {
                RadioButton r = findViewById(id);
                result.setText("You selected: " + r.getText());
            }
        });
    }
}

```

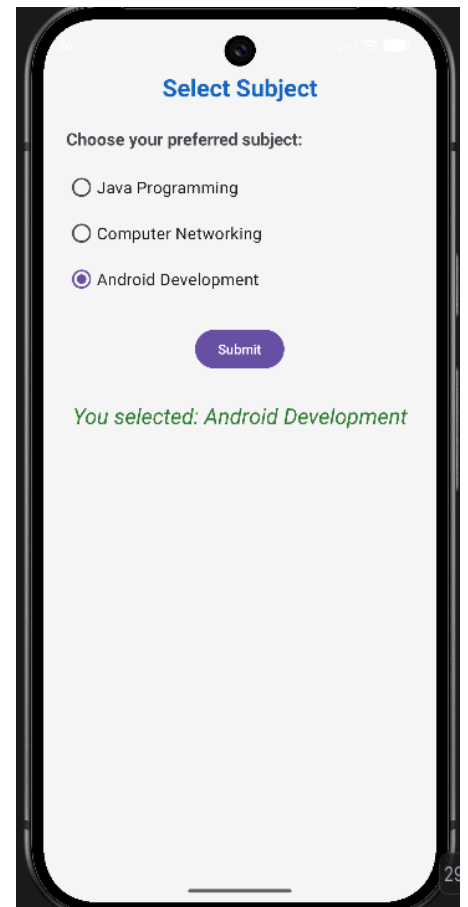
Output:



A mobile application interface titled "Select Subject" in blue. Below the title, it says "Choose your preferred subject:". There are three radio button options: "Java Programming" (selected with a purple dot), "Computer Networking", and "Android Development". A purple "Submit" button is centered below the options. At the bottom, green text reads "You selected: Java Programming". The status bar at the top shows signal strength, Wi-Fi, and battery icons.



A mobile application interface titled "Select Subject" in blue. Below the title, it says "Choose your preferred subject:". There are three radio button options: "Java Programming", "Computer Networking" (selected with a purple dot), and "Android Development". A purple "Submit" button is centered below the options. At the bottom, green text reads "You selected: Computer Networking". The status bar at the top shows signal strength, Wi-Fi, and battery icons.



A mobile application interface titled "Select Subject" in blue. Below the title, it says "Choose your preferred subject:". There are three radio button options: "Java Programming", "Computer Networking", and "Android Development" (selected with a purple dot). A purple "Submit" button is centered below the options. At the bottom, green text reads "You selected: Android Development". The status bar at the top shows signal strength, Wi-Fi, and battery icons.

9. Design an android application for menu.

main_menu.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/settings"
        android:title="Settings" />
    <item
        android:id="@+id/about"
        android:title="About" />
    <item
        android:id="@+id/exit"
        android:title="Exit App" />
</menu>
```

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="MENU APPLICATION"
        android:textSize="25sp"
        android:textStyle="bold"
        android:textColor="#1565C0"
        android:gravity="center"
        android:layout_marginBottom="25dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Tap the three dots : in the top-right corner"
```

```
        android:textSize="17sp"
        android:gravity="center"
        android:layout_marginBottom="30dp"/>
```

```
<TextView
    android:id="@+id/result"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Action Taken: None"
    android:textSize="22sp"
    android:textStyle="bold|italic"
    android:textColor="#E91E63"
    android:gravity="center"/>
```

```
</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.menuapp;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView result;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

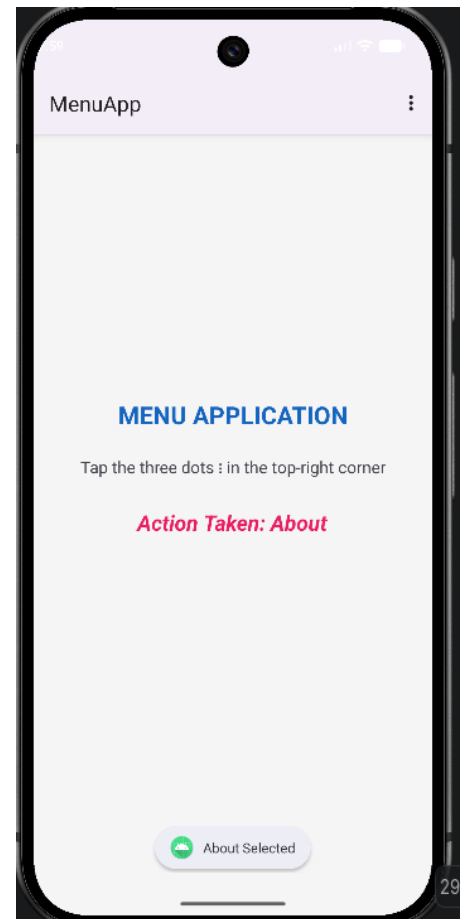
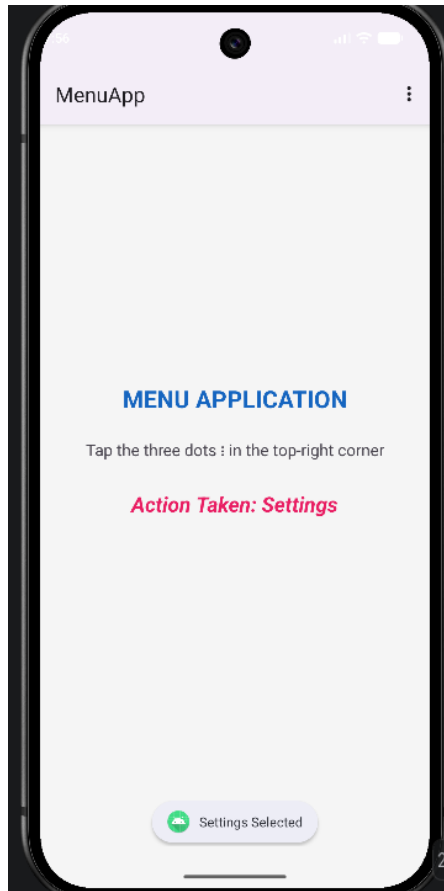
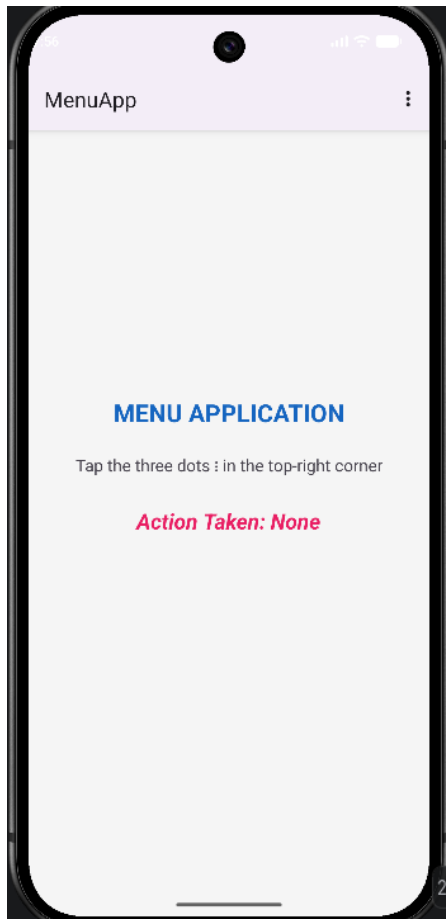
        result = findViewById(R.id.result);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
```

```
public boolean onOptionsItemSelected(MenuItem item) {  
  
    int id = item.getItemId();  
  
    if (id == R.id.settings) {  
  
        result.setText("Action Taken: Settings");  
        Toast.makeText(this, "Settings Selected",  
            Toast.LENGTH_SHORT).show();  
        return true;  
  
    } else if (id == R.id.about) {  
  
        result.setText("Action Taken: About");  
        Toast.makeText(this, "About Selected",  
            Toast.LENGTH_SHORT).show();  
        return true;  
  
    } else if (id == R.id.exit) {  
  
        Toast.makeText(this, "Exiting App",  
            Toast.LENGTH_SHORT).show();  
        finish();  
        return true;  
    }  
  
    return super.onOptionsItemSelected(item);  
}  
}
```

Output:



10. Create a user registration application that stores the user details in a database table.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="55dp"
        android:layout_marginBottom="20dp"
        android:textSize="25dp"
        android:textStyle="bold"
        android:text="Deep Ghinaiya (WD03)" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Registration"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/edit_text_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:inputType="textPersonName"
        android:minHeight="48dp"/>

    <EditText
        android:id="@+id/edit_text_email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:minHeight="48dp"/>

    <EditText
```

```

        android:id="@+id/edit_text_password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:layout_marginBottom="30dp"/>
<Button
    android:id="@+id/button_register"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register User"
    android:onClick="registerUser"
    android:layout_marginBottom="16dp"/>
<Button
    android:id="@+id/button_view_users"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="View All Users (in Logcat)"
    android:onClick="viewAllUsers"/>
</LinearLayout>

```

DatabaseHelper.java

Code:

```

package com.example.practical10;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;

public class DatabaseHelper extends SQLiteOpenHelper {
    private static final String TAG = "DatabaseHelper";
    private static final String DATABASE_NAME = "RegistrationDB";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_USERS = "users";
    private static final String COL_ID = "id";
    private static final String COL_NAME = "name";
    private static final String COL_EMAIL = "email";
    private static final String COL_PASSWORD = "password";
    private static final String CREATE_TABLE_USERS = "CREATE TABLE " + TABLE_USERS
+ "("
    + COL_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
    + COL_NAME + " TEXT, "

```

```
+ COL_EMAIL + " TEXT UNIQUE, "  
+ COL_PASSWORD + " TEXT"  
+ ")";
```

```
public DatabaseHelper(Context context) {  
    super(context, DATABASE_NAME, null, DATABASE_VERSION);  
}
```

```
@Override  
public void onCreate(SQLiteDatabase db) {  
    db.execSQL(CREATE_TABLE_USERS);  
    Log.d(TAG, "Database table created successfully: " + TABLE_USERS);  
}
```

```
@Override  
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {  
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);  
    onCreate(db);  
}
```

```
public boolean addUser(String name, String email, String password) {  
    SQLiteDatabase db = this.getWritableDatabase();  
    ContentValues values = new ContentValues();  
    values.put(COL_NAME, name);  
    values.put(COL_EMAIL, email);  
    values.put(COL_PASSWORD, password);  
    long result = db.insert(TABLE_USERS, null, values);  
    db.close();  
    return result != -1;  
}
```

```
public void getAllUsers() {  
    SQLiteDatabase db = this.getReadableDatabase();  
    String selectQuery = "SELECT * FROM " + TABLE_USERS;  
    Cursor cursor = db.rawQuery(selectQuery, null);  
    if (cursor.moveToFirst()) {  
        do {  
            int id = cursor.getInt(cursor.getColumnIndexOrThrow(COL_ID));  
            String name = cursor.getString(cursor.getColumnIndexOrThrow(COL_NAME));  
            String email = cursor.getString(cursor.getColumnIndexOrThrow(COL_EMAIL));  
            Log.d(TAG, "User ID: " + id + ", Name: " + name + ", Email: " + email);  
        } while (cursor.moveToNext());  
    }  
    cursor.close();  
    db.close();  
}
```

```
}  
}
```

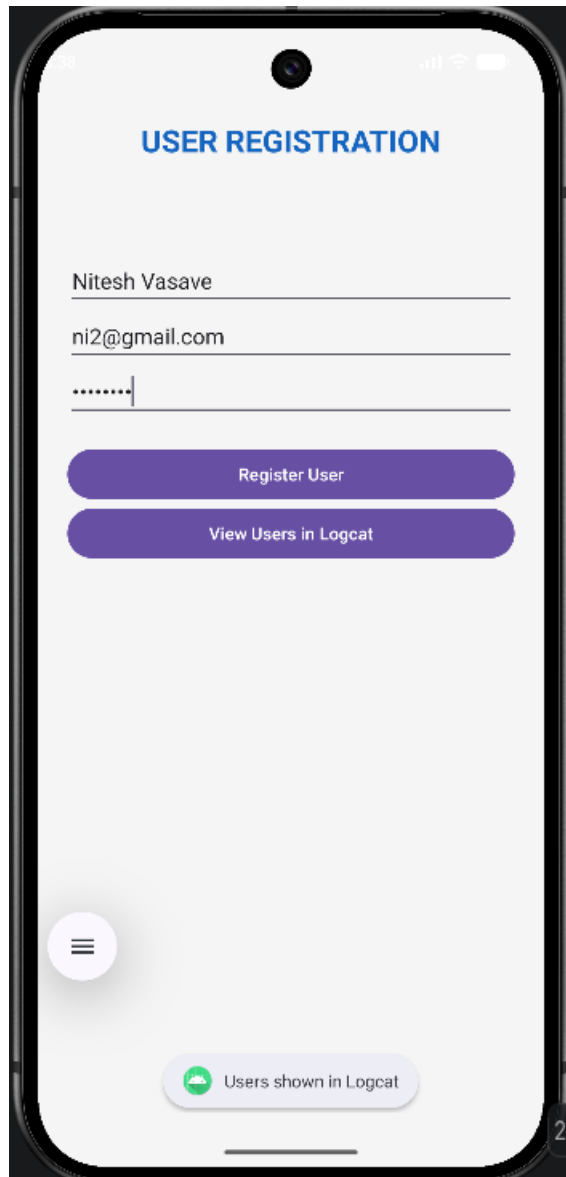
MainActivity.java

Code:

```
package com.example.practical10;  
  
import androidx.appcompat.app.AppCompatActivity;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.EditText;  
import android.widget.Toast;  
public class MainActivity extends AppCompatActivity {  
    private EditText editTextName, editTextEmail, editTextPassword;  
    private DatabaseHelper dbHelper;  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
        editTextName = findViewById(R.id.edit_text_name);  
        editTextEmail = findViewById(R.id.edit_text_email);  
        editTextPassword = findViewById(R.id.edit_text_password);  
        dbHelper = new DatabaseHelper(this);  
    }  
    public void registerUser(View view) {  
        String name = editTextName.getText().toString().trim();  
        String email = editTextEmail.getText().toString().trim();  
        String password = editTextPassword.getText().toString().trim();  
        if (name.isEmpty() || email.isEmpty() || password.isEmpty()) {  
            Toast.makeText(this, "Please fill in all fields.", Toast.LENGTH_SHORT).show();  
            return;  
        }  
        boolean isInserted = dbHelper.addUser(name, email, password);  
        if (isInserted) {  
            Toast.makeText(this, "Registration Successful!", Toast.LENGTH_LONG).show();  
            // Clear inputs after successful registration  
            editTextName.setText("");  
            editTextEmail.setText("");  
            editTextPassword.setText("");  
        } else {  
            Toast.makeText(this, "Registration Failed! Email may already exist.",  
                Toast.LENGTH_LONG).show();  
        }  
    }  
}
```

```
public void viewAllUsers(View view) {  
    dbHelper.getAllUsers();  
    Toast.makeText(this, "User list printed to Logcat (filter by tag 'DatabaseHelper')",  
        Toast.LENGTH_LONG).show();  
}
```

Output:



11. Create an application on to Create, Insert, update, Delete and retrieve operation on the database.

DatabaseHelper.java

Code:

```
package com.example.cruddb;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;

public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "CrudDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(
            "CREATE TABLE users(" +
                "id INTEGER PRIMARY KEY AUTOINCREMENT," +
                "name TEXT," +
                "email TEXT," +
                "password TEXT)"
        );
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS users");
        onCreate(db);
    }

    public boolean insert(String name, String email, String password) {
        SQLiteDatabase db = getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put("name", name);
        values.put("email", email);
        values.put("password", password);

        return db.insert("users", null, values) != -1;
    }
}
```

```

public void view() {
    SQLiteDatabase db = getReadableDatabase();
    Cursor c = db.rawQuery("SELECT * FROM users", null);

    while (c.moveToNext()) {
        Log.d("CRUD", "ID: " + c.getInt(0) + " Name: " + c.getString(1) +
            " Email: " + c.getString(2));
    }
    c.close();
}

public boolean update(String id, String name, String email, String password) {
    SQLiteDatabase db = getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("name", name);
    values.put("email", email);
    values.put("password", password);

    return db.update( "users", values, "id=?", new String[] {id} ) > 0;
}

public boolean delete(String id) {
    SQLiteDatabase db = getWritableDatabase();
    return db.delete("users", "id=?", new String[] {id} ) > 0;
}
}

```

activity_main.xml

Code:

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="80dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="SQLite CRUD Operations"

```

```
android:textSize="25sp"  
android:textStyle="bold"  
android:gravity="center"  
android:layout_marginBottom="20dp"/>
```

```
<EditText  
    android:id="@+id/id"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="ID (for Update/Delete)"  
    android:inputType="number"/>
```

```
<EditText  
    android:id="@+id/name"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Name"/>
```

```
<EditText  
    android:id="@+id/email"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Email"  
    android:inputType="textEmailAddress"/>
```

```
<EditText  
    android:id="@+id/password"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Password"  
    android:inputType="textPassword"/>
```

```
<Button  
    android:id="@+id/insert"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="INSERT"/>
```

```
<Button  
    android:id="@+id/update"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="UPDATE"/>
```

```
<Button  
    android:id="@+id/delete"
```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="DELETE"/>

<Button
    android:id="@+id/view"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="VIEW ALL"/>

</LinearLayout>
</ScrollView>

```

MainActivity.java

Code:

```

package com.example.cruddb;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText id, name, email, password;
    Button insert, update, delete, view;
    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        id = findViewById(R.id.id);
        name = findViewById(R.id.name);
        email = findViewById(R.id.email);
        password = findViewById(R.id.password);

        insert = findViewById(R.id.insert);
        update = findViewById(R.id.update);
        delete = findViewById(R.id.delete);
        view = findViewById(R.id.view);

        db = new DatabaseHelper(this);

        insert.setOnClickListener(v -> {

```

```

        if (name.getText().toString().isEmpty() || email.getText().toString().isEmpty() ||
            password.getText().toString().isEmpty()) {
            Toast.makeText(this, "Fill all fields", Toast.LENGTH_SHORT).show();

        } else if (db.insert( name.getText().toString(), email.getText().toString(),
            password.getText().toString())) {

            Toast.makeText(this, "User Inserted", Toast.LENGTH_SHORT).show();
        }
    });

    update.setOnClickListener(v -> {
        boolean result = db.update(
            id.getText().toString(),
            name.getText().toString(),
            email.getText().toString(),
            password.getText().toString()
        );

        Toast.makeText(this, result ? "User Updated" : "Update Failed",
            Toast.LENGTH_SHORT).show();
    });

    delete.setOnClickListener(v -> {
        boolean result = db.delete(
            id.getText().toString()
        );

        Toast.makeText(this, result ? "User Deleted" : "Delete Failed",
            Toast.LENGTH_SHORT).show();
    });

    view.setOnClickListener(v -> {
        db.view();
        Toast.makeText(this, "Check all data on Logcat",
            Toast.LENGTH_SHORT).show();
    });
}
}

```

Output:

SQLite CRUD Operations

12

Harsh

harsh@gmail.com

INSERT

UPDATE

DELETE

VIEW ALL

☰

🟢 User Inserted

This screenshot shows the 'Insert' screen of the SQLite CRUD Operations app. The title is 'SQLite CRUD Operations'. There are four input fields: a numeric field with '12', a text field with 'Harsh', an email field with 'harsh@gmail.com', and a password field with masked characters '*****'. Below the inputs are four purple buttons: 'INSERT', 'UPDATE', 'DELETE', and 'VIEW ALL'. At the bottom left is a hamburger menu icon (☰), and at the bottom center is a green button with a checkmark icon and the text 'User Inserted'.

SQLite CRUD Operations

1

Karan Vasave

karan11@gmail.com

INSERT

UPDATE

DELETE

VIEW ALL

☰

🟢 User Updated

This screenshot shows the 'Update' screen of the SQLite CRUD Operations app. The title is 'SQLite CRUD Operations'. There are four input fields: a numeric field with '1', a text field with 'Karan Vasave', an email field with 'karan11@gmail.com', and a password field with masked characters '*****'. Below the inputs are four purple buttons: 'INSERT', 'UPDATE', 'DELETE', and 'VIEW ALL'. At the bottom left is a hamburger menu icon (☰), and at the bottom center is a green button with a checkmark icon and the text 'User Updated'.

SQLite CRUD Operations

2

Name

Email

Password

INSERT

UPDATE

DELETE

VIEW ALL

☰

🟢 User Deleted

This screenshot shows the 'Delete' screen of the SQLite CRUD Operations app. The title is 'SQLite CRUD Operations'. There are four input fields: a numeric field with '2', a text field with 'Name', an email field with 'Email', and a password field with 'Password'. Below the inputs are four purple buttons: 'INSERT', 'UPDATE', 'DELETE', and 'VIEW ALL'. At the bottom left is a hamburger menu icon (☰), and at the bottom center is a green button with a checkmark icon and the text 'User Deleted'.

SQLite CRUD Operations

ID (for Update/Delete)

Name

Email

Password

INSERT

UPDATE

DELETE

VIEW ALL

☰

🟢 Check all data on Logcat

This screenshot shows the 'View All' screen of the SQLite CRUD Operations app. The title is 'SQLite CRUD Operations'. There are four input fields: a numeric field with 'ID (for Update/Delete)', a text field with 'Name', an email field with 'Email', and a password field with 'Password'. Below the inputs are four purple buttons: 'INSERT', 'UPDATE', 'DELETE', and 'VIEW ALL'. At the bottom left is a hamburger menu icon (☰), and at the bottom center is a green button with a checkmark icon and the text 'Check all data on Logcat'.

12. Create an application on with three op on buttons, on selecting a button color of the screen will change.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center_horizontal"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CHANGE SCREEN COLOR"
        android:textSize="25sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="100dp"
        android:layout_marginBottom="30dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select a Color:"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="15dp"/>

    <RadioGroup
        android:id="@+id/radio_group_colors"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <RadioButton
            android:id="@+id/radio_red"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Red"
            android:textSize="17sp"/>
```

```

<RadioButton
    android:id="@+id/radio_green"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Green"
    android:textSize="17sp"/>

<RadioButton
    android:id="@+id/radio_blue"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Blue"
    android:textSize="17sp"/>

</RadioGroup>
</LinearLayout>

```

MainActivity.java

Code:

```

package com.example.colorapp;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.LinearLayout;
import android.widget.RadioGroup;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    LinearLayout mainLayout;
    RadioGroup colors;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        mainLayout = findViewById(R.id.main_layout);
        colors = findViewById(R.id.radio_group_colors);

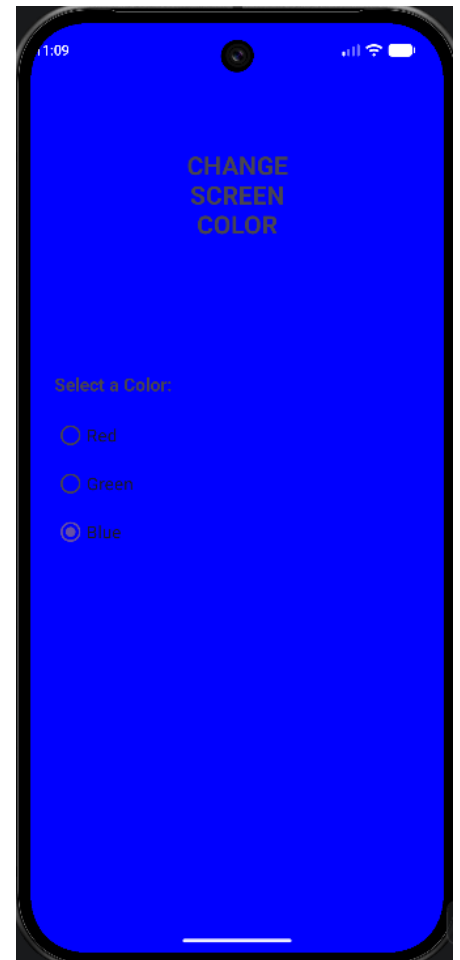
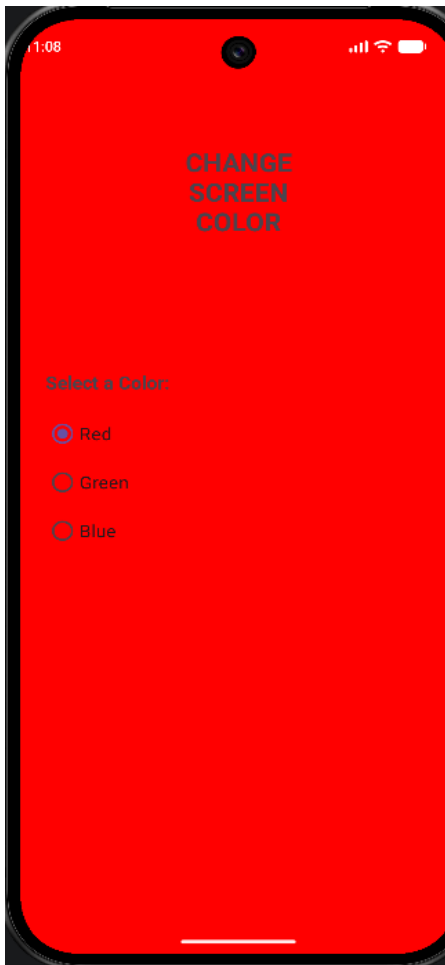
        colors.setOnCheckedChangeListener((group, id) -> {

            if (id == R.id.radio_red) {
                mainLayout.setBackgroundColor(Color.RED);
            }
        });
    }
}

```

```
    } else if (id == R.id.radio_green) {  
        mainLayout.setBackgroundColor(Color.GREEN);  
  
    } else if (id == R.id.radio_blue) {  
        mainLayout.setBackgroundColor(Color.BLUE);  
    }  
    });  
}  
}
```

Output:



13. Create spinner with strings taken from resource folder (res >> value folder) and on changing the spinner value, Image will change.

strings.xml

Code:

```
<resources>
    <string name="app_name">Practical 13</string>
    <string name="action_settings">Settings</string>

    <string-array name="planet_names">
        <item>Earth</item>
        <item>Mars</item>
        <item>Jupiter</item>
        <item>Saturn</item>
    </string-array>
</resources>
```

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center_horizontal"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="55dp"
        android:layout_marginBottom="20dp"
        android:textSize="25dp"
        android:textStyle="bold"
        android:text="Deep Ghinaiya (WD03)" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select a Planet:"
```

```

        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp"/>
<Spinner
    android:id="@+id/spinner_planets"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:layout_marginBottom="32dp"/>
<ImageView
    android:id="@+id/image_view_planet"
    android:layout_width="200dp"
    android:layout_height="200dp"
    android:scaleType="fitCenter"
    android:src="@drawable/earth"
    android:contentDescription="Selected Planet Image" />
</LinearLayout>

```

MainActivity.java

Code:

```

package com.example.practical13;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.ImageView;
import android.widget.Spinner;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    private Spinner spinnerPlanets;
    private ImageView imageViewPlanet;
    private String[] planetNames; // Array to hold the string resource values
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        spinnerPlanets = findViewById(R.id.spinner_planets);
        imageViewPlanet = findViewById(R.id.image_view_planet);
        planetNames = getResources().getStringArray(R.array.planet_names);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_spinner_item, // Default layout for spinner item
            planetNames // The string array data
        );
        spinnerPlanets.setAdapter(adapter);
    }
}

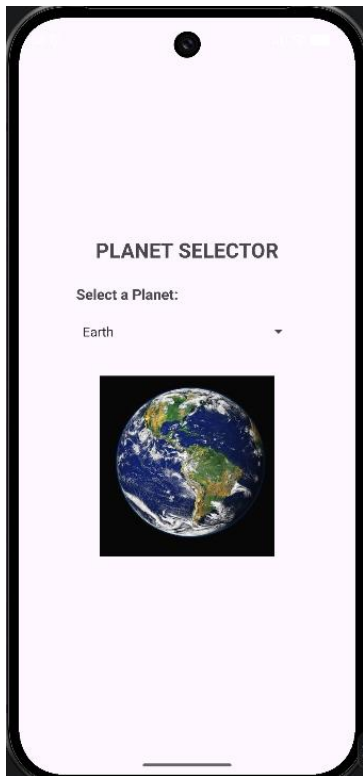
```

```

);
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
spinnerPlanets.setAdapter(adapter);
spinnerPlanets.setOnItemClickListener(new AdapterView.OnItemClickListener() {
    @Override
    public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
// 'position' is the index of the newly selected item (0 for Earth, 1 for Mars, etc.)
        changeImage(position);
    }
    @Override
    public void onNothingSelected(AdapterView<?> parent) {
// Can be ignored for this practical
    }
});
changeImage(0);
}
private void changeImage(int position) {
    String selectedPlanet = planetNames[position].toLowerCase();
    int imageResourceId;
    if (selectedPlanet.equals("earth")) {
        imageResourceId = R.drawable.earth;
    } else if (selectedPlanet.equals("mars")) {
        imageResourceId = R.drawable.mars;
    } else if (selectedPlanet.equals("jupiter")) {
        imageResourceId = R.drawable.jupiter;
    } else if (selectedPlanet.equals("saturn")) {
        imageResourceId = R.drawable.saturn;
    } else {
        imageResourceId = R.drawable.earth;
    }
    imageViewPlanet.setImageResource(imageResourceId);
    Toast.makeText(this, "Displaying " + planetNames[position],
Toast.LENGTH_SHORT).show();
}
}

```

Output:



14. Create an application with login module.

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="30dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="USER LOGIN"
        android:textSize="28sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="100dp"
        android:layout_marginBottom="35dp"/>

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:inputType="text"
        android:layout_marginBottom="15dp"/>

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="25dp"/>

    <Button
        android:id="@+id/login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="LOGIN"/>

    <TextView
```

```
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Enter username and password"
        android:textSize="18sp"
        android:gravity="center"
        android:layout_marginTop="25dp"/>
</LinearLayout>
```

MainActivity.java

Code:

```
package com.example.login;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText username, password;
    Button login;
    TextView result;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        login = findViewById(R.id.login);
        result = findViewById(R.id.result);

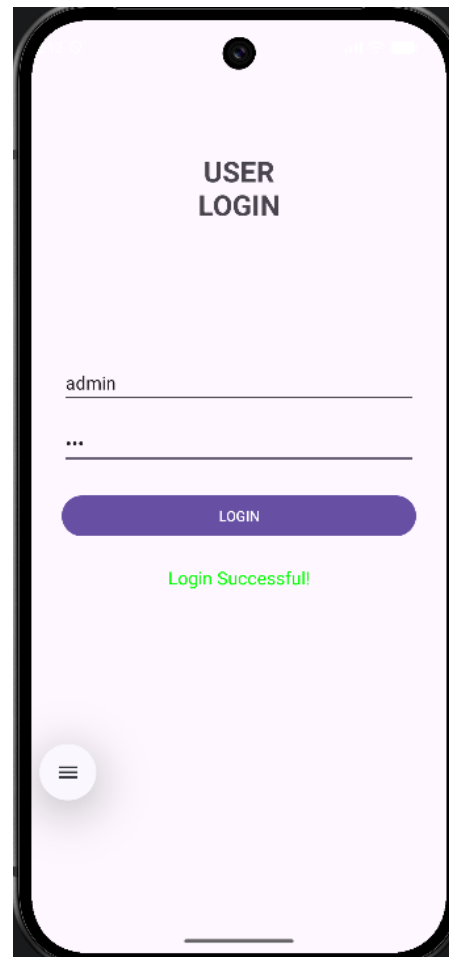
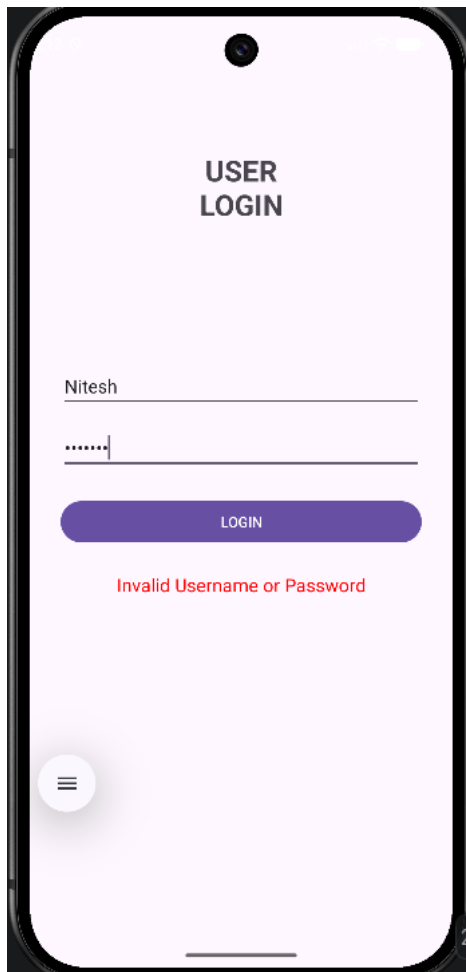
        login.setOnClickListener(v -> {
            String user = username.getText().toString().trim();
            String pass = password.getText().toString().trim();

            if (user.isEmpty() || pass.isEmpty()) {
                result.setText("Please enter all fields");
            } else if (user.equals("admin") &&
                pass.equals("123")) {

                result.setText("Login Successful!");
            }
        });
    }
}
```

```
        result.setTextColor(Color.GREEN);  
    } else {  
        result.setText("Invalid Username or Password");  
        result.setTextColor(Color.RED);  
    }  
    });  
}  
}
```

Output:



15. Create a list of all courses in your college and on selecting a particular course teacher charge of that course should appear at the bottom of the screen.

strings.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>

    <string name="app_name">Course App</string>
    <string name="course_list_title">COURSE LIST</string>
    <string name="select_course_label">Select a Course:</string>
    <string name="teacher_in_charge_default">Teacher-in-Charge: Not Selected</string>
    <string name="teacher_in_charge_format">Teacher-in-Charge: %1$s</string>

    <string-array name="course_names">
        <item>MCA-301: Advanced Java</item>
        <item>MCA-302: Android Programming</item>
        <item>MCA-303: Cloud Computing</item>
        <item>MCA-304: Database Management</item>
    </string-array>

    <string-array name="teacher_names">
        <item>Prof. Anamika Dixit</item>
        <item>Prof. Aishwarya Borse</item>
        <item>Prof. Gautami Upadhyay</item>
        <item>Dr. Manojkumar Langote</item>
    </string-array>
</resources>
```

activity_main.xml

Code:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:background="#FFFFFF">

    <TextView
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:text="@string/course_list_title"
        android:textSize="28sp"
        android:textStyle="bold"
        android:textColor="#000000"
        android:gravity="center"
        android:layout_marginTop="40dp"
        android:layout_marginBottom="30dp"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/select_course_label"
    android:textSize="20sp"
    android:textStyle="bold"
    android:textColor="#000000"
    android:layout_marginBottom="15dp"/>

<Spinner
    android:id="@+id/spinner_courses"
    android:layout_width="match_parent"
    android:layout_height="55dp"/>

<Space
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"/>

<TextView
    android:id="@+id/text_view_teacher"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/teacher_in_charge_default"
    android:textSize="20sp"
    android:textStyle="bold"
    android:textColor="#3F51B5"
    android:gravity="center"
    android:padding="15dp"
    android:layout_marginBottom="20dp"/>
</LinearLayout>

```

MainActivity.java

Code:

```

package com.example.practical15;

import androidx.appcompat.app.AppCompatActivity;

```

```
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    private Spinner spinnerCourses;
    private TextView textViewTeacher;
    private String[] teacherNames; // Array to hold teacher names
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        spinnerCourses = findViewById(R.id.spinner_courses);
        textViewTeacher = findViewById(R.id.text_view_teacher);
        String[] courseNames = getResources().getStringArray(R.array.course_names);
        teacherNames = getResources().getStringArray(R.array.teacher_names);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_spinner_item, // Default layout for spinner item
            courseNames // The course names array
        );
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        spinnerCourses.setAdapter(adapter);
        spinnerCourses.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
            @Override
            public void onItemSelected(AdapterView<?> parent, View view, int position, long id) {
                displayTeacherInCharge(position);
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {
            }
        });
    }
    private void displayTeacherInCharge(int position) {
        String teacher = teacherNames[position];
        String result = "Teacher-in-Charge: " + teacher;
        textViewTeacher.setText(result);
        Toast.makeText(this, "Course selected: " + teacher, Toast.LENGTH_SHORT).show();
    }
}
```

Output:



